

— CLASS PROJECT · ONE TERM · ONE V100

You cannot reproduce their setup. Build one that measures what you can run.

Pick one benchmark paper. Its models and its infrastructure are out of reach on a V100 — that is the starting condition, not the failure. Your job is an instrument that makes *small open-weight models* meaningfully assessable on that benchmark, and a gap read honestly against the numbers the paper reports.

Setup · 60%

A measurement small models can be judged by, and a reader can re-run.

Findings · 40%

What your numbers show, and what the gap to the paper does and does not license.

Harness · optional

Improvements to the evaluation machinery itself, demonstrated against the original.

Suggested benchmark: IFEval — 25 verifiable instruction types over 541 prompts, checked by program, so no judge to validate and no API bill [Zhou et al. 2023]. Any substitute must be exactly checkable — and if you want a cross-language finding, it must also carry more than one language. Companion to *Evaluating AI Systems*.

What makes a small model meaningfully assessable.

REQUIREMENT	WHAT IT MEANS, CONCRETELY
The paper’s items and its scorer	Same item list, same metric definitions, the authors’ scoring code. If you reimplement, validate against theirs on a subset and report the agreement.
A documented prompt path	The template you apply and the <i>exact final string</i> the model sees. Print it. This is the single largest source of spurious difference.
Decoding pinned	Greedy as the baseline. Sampling only as a deliberate variance experiment, never as the default nobody wrote down.
A model that actually fits	Memory budget before checkpoint choice: fp16 weights $\approx$ 2 bytes/parameter, plus KV cache. Volta gives you fp16 tensor cores, no bf16, and no FlashAttention-2.
Repeats, and a spread	Several runs at one configuration, then several meaning-preserving rewordings. A gap smaller than your own spread is not a gap [ReliableEval, 2025].
Everything pinned and logged	Model revision, tokenizer, precision, seed, library and scorer versions, plus the raw generations and whatever your parser discarded.

The test of a good setup: someone else can run it and get your number, and can tell from the artifacts alone what was measured. That is worth more marks than any result, because without it no result means anything.

Read the gap against the paper, and bound what it means.

REPORT
All four IFEval metrics, with intervals — and say which one you headline
A per-instruction-type breakdown: which of the 25 your model fails
Response-length distribution and truncation rate
GPU-hours per full pass
The gap to the paper’s reported figures: size and direction

What the gap is confounded with

Different model, different scale, different precision, different template, different decoding. You changed five things at once — **name them** rather than attributing the difference to size.

What it still tells you

Which instruction types collapse at small scale, how wide the spread is, and whether the benchmark discriminates at all in this range. That is a finding the paper does not contain.

Write the claim as a sentence you can defend: “*Model M at revision R, fp16 on a V100, greedy, template C, scored X% prompt-level strict (range [a, b] over n runs); the paper reports Y% for a different and larger model. The difference is confounded with ...*” If you cannot fill every blank, the finding is not yet a finding.

Sixty on the instrument, forty on what it showed.

COMPONENT	WEIGHT	EARNs FULL MARKS WHEN
The setup	60%	A stranger can re-run it: pinned versions, exact prompts, the scorer, raw generations, one-command entry point — and repeats with a reported spread
The findings	40%	All four metrics with intervals, a per-type breakdown, the gap stated with its confounds named, and an explicit residual. A result that departs from the paper’s picture counts fully.
Harness improvements	optional	Anything that makes the evaluation machinery better, demonstrated against the original: stronger output parsing, a prompt-template ablation, a contamination probe, throughput work that buys more repeats, or a fix to a scorer edge case

A finding may depart from the paper’s picture

Well evidenced, that scores full marks. For example: *“Unlike the results from frontier models reported in the paper, smaller models struggle to perform basic reasoning, and exhibit greater performance discrepancies across languages.”* A **cross-language claim needs a benchmark with language coverage** — decide that when you choose the paper, not afterwards.

The rule for harness work

An improvement must be **shown to be one**: measure the original, measure yours, and report both. An unmeasured improvement is a preference.

Note what is **not** marked: agreement with the published number. You are building an instrument and reporting what it read.